Shicheng Fan

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EDUCATION

University of Illinois Chicago (UIC) Aug. 2025 – Present
Ph.D. in Computer Science Chicago, IL

- Research Areas: Causality, RL, LLM Agents & Alignment, AI for Science

Zhejiang University, Chu Kochen Honors College Sept. 2021 – June 2025
B.Eng. in Automation (Control and Robotics) Hangzhou, China

- GPA: 3.98/4.0, Rank: 6/121 (Top 5%), Outstanding Graduate

PUBLICATIONS & PREPRINTS

[1] **Shicheng Fan**, Kun Zhang, Lu Cheng. "TRACE: Trajectory Recovery for Continuous Mechanism Evolution in Causal Representation Learning." **ICML 2026**

[2] Hanrong Zhang*, **Shicheng Fan***, ..., Philip S. Yu. "EvoSkills: Self-Evolving Agent Skills via Co-Evolutionary Verification." Under Review

RESEARCH EXPERIENCE

TRACE: Trajectory Recovery for Continuous Mechanism Evolution in CRL | *1st Author* ICML 2026

- Identified continuous mechanism evolution as an open problem in CRL; proposed the first framework modeling intermediate mechanisms as convex combinations of atomic mechanisms; proved joint identifiability via nonlinear ICA with finite-sample error bounds
- Designed a Mixture-of-Experts architecture recovering mechanism trajectories; validated on synthetic and real-world data (UAVDT, CMU MoCap), achieving 0.99 correlation and outperforming discrete-switching baselines

EvoSkills: Self-Evolving Agent Skills via Co-Evolutionary Verification | *Co-1st Author* Under Review

- Proposed EvoSkills, a co-evolutionary framework for autonomous multi-file skill generation; designed information-isolated verification (Surrogate Verifier + GT Oracle) to eliminate confirmation bias in self-improvement loops
- Improved pass rate by ∼40% on Claude Code and Codex on SkillsBench (87 tasks, 20+ domains); evolved skills transfer to 5 additional LLMs with 30–40% gains

Knowledge-Enhanced RL for LLM Factuality Alignment | *Co-1st Author* In Preparation

- Designed an end-to-end LLM post-training pipeline (cold-start SFT + GRPO RL) with multi-dimensional rewards; post-trained DeepSeek-R1-7B/14B with LoRA on A100 clusters, achieving SOTA factuality on TriviaQA and SimpleQA
- Proposed CoVer, a plug-and-play corpus-verified factuality reward: extracted knowledge triples via a 0.5B extractor, verified co-occurrence against Dolma corpus (∼25TB indexed), providing an API-free reward for any RL framework

MOSAIC: Module Discovery via Sparse Identifiable Causal Learning | *1st Author* In Preparation

- Formulated a new CRL setting for scientific time series; proposed MOSAIC, combining temporal nonstationarity with mechanism sparsity (additive decoder + group lasso) for bidirectional interpretability; proved module identifiability and finite-sample support recovery bounds
- Validated across 6 domains (RNA dynamics, solar flare, tokamak, EEG, climate, epidemiology); achieved F1 = 1.0 module recovery on synthetic benchmarks and 99.7% regime accuracy on RNA